

The Arrow from Constraint

On Temporal Asymmetry in Memory Topology

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Abstract

In earlier work, time was modeled as an emergent ordering relation induced by memory topology rather than as a primitive external dimension. That formulation captured succession, but left a sharper question unresolved: under what conditions does a constraint topology generate a genuine temporal arrow rather than a merely reversible reachability relation? This note addresses that gap. I argue that ordering alone is insufficient. A temporal arrow appears only when the constraint structure itself is directionally biased, whether through accumulative asymmetry, topological inaccessibility, or differential cost of reversal. The aim is not to derive a full physical theory of time, but to sharpen the earlier phenomenological model by identifying the minimal structural conditions under which temporal asymmetry emerges.

1. Introduction

In earlier work, I proposed that time emerges from memory topology as the ordering induced by admissible state transitions. Under that framework, memory was modeled not as stored content but as a constraint structure shaping what states could follow one another. Temporal order therefore arose from reachability rather than from motion along an external time coordinate.

That formulation captured succession, but it left a sharper question unresolved: what distinguishes a genuine temporal arrow from a merely reversible reachability relation? If the topology of admissible transitions is symmetric, then order may exist without asymmetry. A system may exhibit sequence, adjacency, or even cyclic recurrence without yielding anything recognizably like directed temporal passage.

The present note addresses that gap. I argue that memory topology alone does not generate a temporal arrow unless the constraint structure itself is directionally biased. Ordering is necessary, but not sufficient. A genuine arrow appears only when the topology remembers direction through accumulation, inaccessible reversal, or differential cost of return.

The scope of this paper remains modest. It does not attempt to solve the thermodynamic arrow of time, revise physical cosmology, or derive temporal asymmetry from fundamental physics. Its aim is narrower: to refine the earlier memory-topology model by identifying the minimal structural conditions under which temporal asymmetry can emerge within a constrained system.

2. Reachability Is Not Enough

The memory-topology framework defines temporal order as reachability: $s_i \prec s_j$ if and only if there exists a path of admissible transitions from s_i to s_j under M . This induces a preorder on states, capturing succession and adjacency without invoking an external clock. But reachability alone does

not yet yield a temporal arrow. It provides connectivity and sequence, while remaining agnostic about direction unless asymmetry is present.

Consider a symmetric topology in which $(s_i, s_j) \in M$ implies $(s_j, s_i) \in M$. In that case, the induced relation is effectively bidirectional: states remain mutually accessible, yielding loops or equivalence classes rather than irreversible progression. A simple toy example is a cycle graph, where states form a closed loop with uniform bidirectional edges. Traversal is possible in either direction, producing cyclic sequence but no intrinsic forward bias. The system exhibits adjacency and periodicity, but not the unidirectional flow characteristic of temporal experience.

A similar issue appears in a densely connected symmetric graph. If nearly all states are mutually adjacent, the system supports many possible paths but privileges none of them as temporally primary. In such a case, order exists only in the weak sense of accessibility, not in the stronger sense of directed passage.

The conclusion is therefore simple: memory topology may generate order without yet generating an arrow. To obtain a genuine temporal asymmetry, the constraint structure itself must retain or impose directional bias. It is to those conditions that we now turn.

3. Conditions for a Temporal Arrow

If reachability alone is insufficient, then a genuine temporal arrow requires more than admissible sequence. It requires that the constraint structure itself be directionally biased. The question is not whether states can be ordered, but what makes one direction of that ordering functionally privileged over the other.

Three minimal conditions appear especially relevant.

3.1 Accumulative Asymmetry

A temporal arrow emerges when later states retain structural consequences of earlier transitions, while earlier states do not equivalently retain access to later ones. In this case, the topology is not merely traversed; it is altered or sedimented by traversal. History is not stored as explicit content, but is carried forward as accumulated constraint.

This introduces a directional bias: the future remains shaped by prior realizations, while the past is not symmetrically re-opened by what follows. The system therefore acquires path dependence. Ordered succession becomes temporally asymmetric when transition itself modifies the accessibility structure of subsequent states.

3.2 Topological Inaccessibility

A second route to temporal asymmetry arises when reverse paths are absent from the topology altogether. In such cases, some transitions are admissible in one direction but not in the other. The resulting asymmetry is not energetic or statistical, but structural: the topology itself forbids return.

This produces a stronger arrow than simple ordering. The system does not merely tend to move forward; it lacks the topological capacity to retrace certain paths. Temporal direction then appears

as an effect of missing reversibility within the constraint manifold.

3.3 Reversal Cost

A third possibility is that reverse transitions remain formally possible, but carry a much higher structural or energetic cost than forward ones. Here the topology does not strictly forbid reversal, but strongly suppresses it. The arrow emerges not from impossibility, but from asymmetry of accessibility.

This case is especially important because many real systems are not absolutely irreversible. Instead, they exhibit effective irreversibility: return paths exist in principle, but are so costly, unstable, or combinatorially unlikely that the system behaves directionally. In such cases, temporal asymmetry is a property of weighted constraint rather than binary permission.

3.4 Summary

These three cases are not mutually exclusive. A given system may exhibit more than one at once. But they clarify the central point: temporal order becomes a temporal arrow only when the constraint structure remembers direction—whether through accumulated deformation, inaccessible reversal, or differential cost of return.

What emerges from memory topology, then, is not automatically time in the full directional sense. It is time's arrow only when the topology is biased.

4. Memory as Sediment

The preceding conditions suggest that memory cannot be understood merely as stored content or retrievable representation. If memory topology is to generate a genuine temporal arrow, then memory must function not only as constraint, but as accumulated deformation of accessibility.

This is the sense in which memory behaves more like sediment than archive.

An archive preserves the past as content available for lookup. Sediment, by contrast, preserves the past by altering the structure through which subsequent movement occurs. Its significance lies not in what is explicitly stored, but in how it changes what can happen next. Under this view, memory does not need to contain a record in order to exert continuity. It shapes the future by leaving behind a topology marked by prior traversal.

This distinction clarifies why memory can generate direction without requiring explicit recall. A system may move forward under constraints inherited from prior states even when no previous state is symbolically represented. The past survives not as content but as bias.

Such a formulation also helps explain why temporal experience often feels asymmetric even in the absence of reflection. One does not need to remember every prior state in order to experience forward passage. It is enough that prior transitions have altered the accessibility of later ones. Time, in this sense, does not arise from replaying history, but from moving within a structure that history has already shaped.

This notion of sedimented constraint also provides a natural bridge to lived examples. Dreams, trauma, and narrative continuity all suggest cases in which temporal experience is carried by altered accessibility rather than by explicit retrieval. The system moves as it does because its topology has already been marked.

Remark 4.1. *Memory is not storage but accumulated deformation of accessibility.*

If this is right, then the arrow of time does not emerge simply because states can be ordered, but because the constraint structure itself bears the asymmetric imprint of prior traversal.

5. Illustrative Cases

The preceding framework remains abstract unless it can be related back to recognizable forms of temporal experience. Three cases are especially useful here: dreaming, trauma, and narrative reframing. These do not prove the model, but they help clarify how directional bias in a constraint topology may appear phenomenologically.

5.1 Dreaming

Dreaming provides a natural example of ordered succession without external clocks or sensory input. Dreams often exhibit sequence, causality, and duration despite the absence of direct reference to objective time. This suggests that temporal experience can be generated internally by the structure of state transitions alone.

Within the present framework, dreaming can be understood as a case of accumulative asymmetry. Each dream-state conditions the accessibility of what follows, producing a local order that is carried forward even when no external timeline is present. The arrow here is weak but real: the dream unfolds through constrained succession rather than through explicit clock time.

5.2 Trauma

Trauma provides a more severe example of topological distortion. In traumatically conditioned systems, certain transitions may become inaccessible, cyclic, or disproportionately weighted. This can produce the familiar experience of temporal disruption: the sense that one cannot fully leave a prior state behind, or that the future remains trapped by the structure of an earlier event.

In the present terms, trauma may be understood as a deformation of the accessibility manifold. The issue is not simply that painful content is remembered, but that the topology itself has been altered such that certain paths remain blocked while others become compulsively recurrent. Temporal distortion then appears as a consequence of constrained asymmetry.

5.3 Narrative Reframing

Narrative reframing provides a case in which the topology of access is altered without erasing the past. A system does not become temporally coherent by deleting prior states, but by changing how those states constrain future transitions. Reframing modifies the shape of admissible continuation.

This makes it a useful example of reversal cost and topological repair. The past remains present, but the cost of moving through it changes. What was formerly inaccessible or cyclic may become traversable again. In this sense, narrative intervention does not overwrite memory; it redistributes accessibility.

5.4 Summary

These cases differ in scale and mechanism, but they share one feature: temporal experience is shaped less by explicit recall than by the way prior states alter the topology of what can happen next. Dreams show internally generated order, trauma shows deformation of access, and reframing shows that the arrow can be softened or redirected by changing the constraint structure itself.

6. Minimal Formal Proposal

The preceding discussion suggests that temporal asymmetry requires more than ordered reachability. It requires that the constraint topology possess a directional bias. To state this minimally, let

$$A(M)$$

denote an asymmetry functional on the memory topology M , where $A(M)$ measures the degree to which admissible transitions are directionally imbalanced.

We do not attempt here to specify a unique form for $A(M)$. The present aim is only to distinguish two limiting cases:

$$A(M) = 0 \quad \Rightarrow \quad \text{ordering without intrinsic arrow,} \tag{1}$$

$$A(M) > 0 \quad \Rightarrow \quad \text{ordering with directional bias.} \tag{2}$$

In the first case, the system may exhibit sequence or reachability but no privileged temporal direction. In the second, the topology itself favors one direction through accumulation, inaccessible reversal, or differential cost. The arrow of time then appears not as an external parameter, but as a property of biased constraint structure.

A more complete treatment would require explicit models of $A(M)$, possibly in terms of directional reachability volumes, weighted transition asymmetries, or evolving accessibility metrics. That work is left for future investigation. For the purposes of this note, the important point is simpler: a memory topology generates a temporal arrow only when its constraint structure is not merely ordered, but directionally marked.

7. Scope and Limits

This note does not attempt to solve the thermodynamic arrow of time, derive cosmological time asymmetry, or revise the treatment of time in fundamental physics. Its scope remains phenomenological and structural. The question addressed here is not why the universe as a whole evolves irreversibly, but what conditions must hold for a constrained system to exhibit a genuine arrow of temporal experience.

Nor does this paper claim that all temporal asymmetry reduces to memory in the ordinary psychological sense. The point is broader: any system whose accessibility structure accumulates directional bias may exhibit an arrow. Memory topology is one instance of that general principle.

8. Conclusion

Earlier work proposed that time emerges from memory topology as the ordering induced by admissible state transitions. The present note sharpens that claim. Ordering alone is not yet a temporal arrow. A genuine arrow appears only when the constraint structure itself is directionally biased—through accumulated deformation, topological inaccessibility, or differential cost of reversal.

The result is a more precise statement of the original intuition: time does not emerge simply because states can be sequenced, but because the topology of access retains asymmetry. In this sense, memory does not function as storage but as sedimented constraint.

This revision matters beyond the original time model. If temporal asymmetry depends on biased constraint structure, then any broader attempt to relate emergent order to rendered geometry must account not only for topology, but for the way topology remembers direction. That question lies at the boundary of the present note and the next stage of the work.

References

1. L. Yarbrough, *Time as Emergent from Memory Topology*, preprint, 2025.
2. L. Yarbrough, *Constraint Before Dimension: A Minimal Bridge Between Emergent Time and Rendered Space*, preprint, 2026.